

PLANETARY MOTION

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THE PLANETS IN THE SOLAR SYSTEM are much closer to Earth than are the stars, so as they orbit the Sun they appear to wander across the starry background. This sky motion is influenced by Earth's own solar orbit, which changes the point

of view of Earth-bound observers. The planets closest to Earth move around on the celestial sphere more rapidly than the more distant planets; this is partly due to perspective and partly because the closer a planet is to the Sun, the faster is its orbital speed.

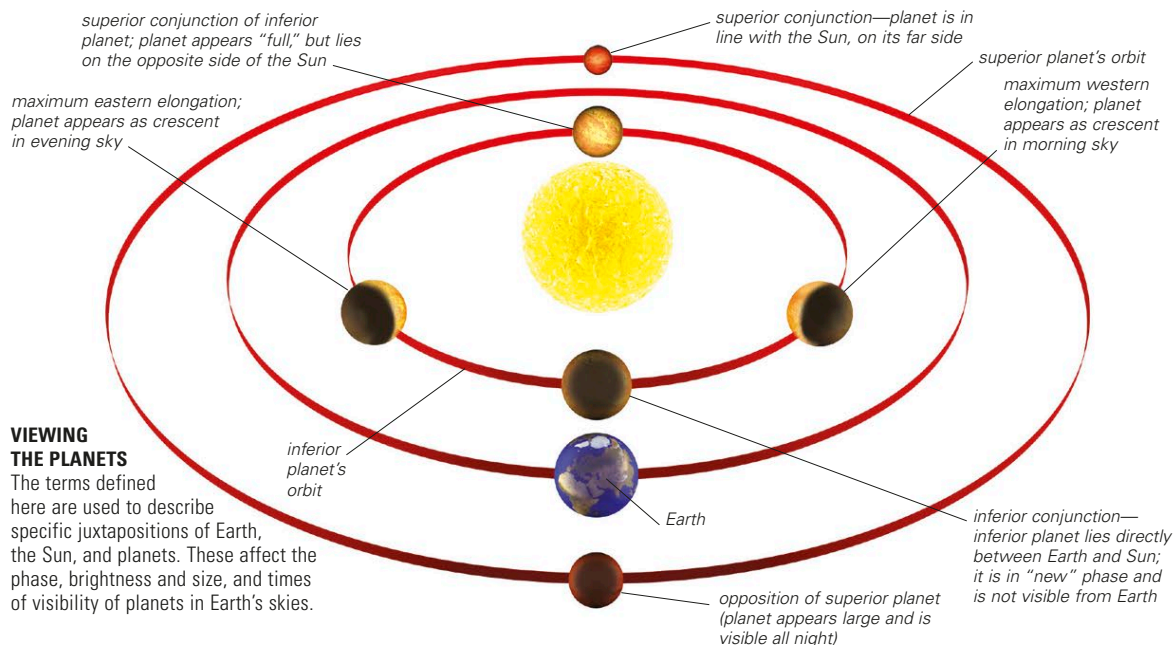
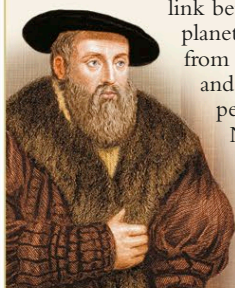
INFERIOR AND SUPERIOR PLANETS

In terms of their motions in the sky as seen from Earth, the planets are divided into two groups. The inferior planets, Mercury and Venus, are those that orbit closer to the Sun than does Earth. They never move far from the Sun on the celestial sphere—the greatest angle by which the planets stray from the Sun (called their maximum elongation) is 28° for Mercury and 45° for Venus. Because they are close to Earth and orbiting quickly, both planets move rapidly against the background stars. They also display phases, like the Moon's (see p.66), because there is some variation in the angle between Earth, the planet, and the Sun. All the other planets, from Mars outward, are called superior planets. These are not “tied” to the Sun on the celestial sphere, and so can be seen in the middle of the night. Apart from Mars, the superior planets are too far from Earth to display clear phases, and they move slowly on the celestial sphere—the farther they are from the Sun, the slower their movement.

JOHANNES KEPLER

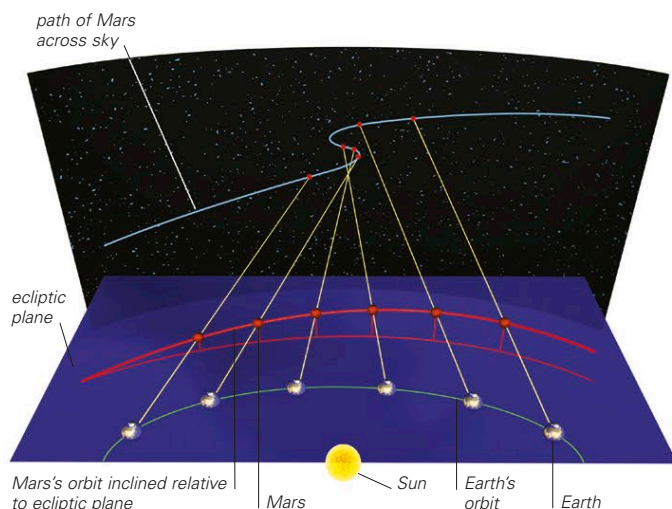
The German astronomer Johannes Kepler (1571–1630) discovered the laws of planetary motion. His first law states that planets orbit the Sun in elliptical paths. The next states that the closer a planet comes to the Sun, the faster it moves, while his third law describes the link between a planet's distance from the Sun and its orbital period.

Newton used Kepler's laws to formulate his theory of gravity.



VIEWING THE PLANETS

The terms defined here are used to describe specific juxtapositions of Earth, the Sun, and planets. These affect the phase, brightness and size, and times of visibility of planets in Earth's skies.



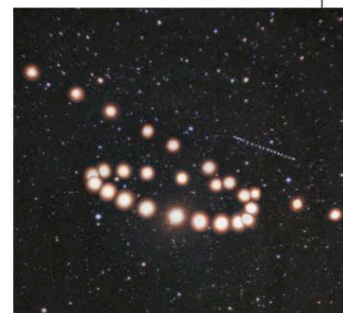
RETROGRADE MOTION

The planets generally move through the sky from west to east against the background of stars, night by night. However, periodically, a planet moves from east to west for a short time—a phenomenon called retrograde motion. Retrograde motion is an effect of changing perspective. Superior planets such as Mars show retrograde motion when Earth “overtakes” the other planet at opposition (when Earth moves between the superior planet and the Sun). The inferior planets Mercury and Venus show retrograde

ZIGZAG ON THE SKY

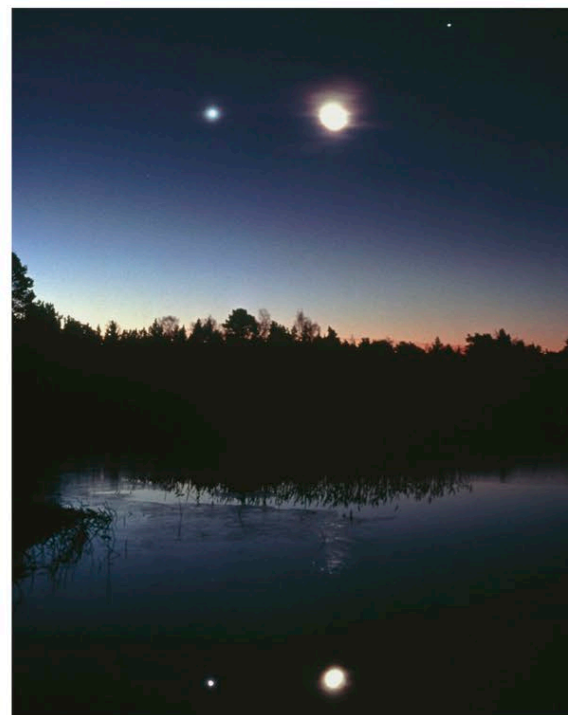
In retrograde motion, a planet may perform a loop or a zigzag on the sky, depending on the angle of its orbit relative to Earth's.

motion on either side of inferior conjunction. They overtake Earth as they pass between Earth and the Sun.



MARTIAN LOOP-THE-LOOP

This composite of photographs taken over several months shows a retrograde loop in Mars's motion against the background stars. The additional short dotted line is produced by Uranus.



ALWAYS NEAR THE SUN

The Moon and Venus appear close together here in the dawn sky. Venus is only ever visible in the eastern sky for up to a few hours before dawn, or in the western sky after dusk—it is never seen in the middle of the night. This is because it orbits closer to the Sun than Earth and so never strays far from the Sun in the sky.